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Confirmatory Factor Analysis With Categorical Indicators

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```
library(here)
library(lavaan)
library(dplyr)
library(modelsummary)
```

The data come from this paper: https://doi.org/10.1111/desc.12918. The authors kindly shared the data on OSF: https://osf.io/hq4xy/overview.

```
# Download data from OSF
if (!dir.exists(here("data"))) {
    dir.create(here("data"))
}
# Raw items
tiego_raw_name <- here("data", "tiego_raw.csv")
if (!file.exists(tiego_raw_name)) {
    download.file(
        "https://osf.io/download/mtnqj/",
        destfile = tiego_raw_name
    )
}
```

Load Data

```
tiego_raw <- read.csv(tiego_raw_name)
# Notice that the minimum is -999 for many variables, so -999 is likely a
missing code
# Treat -999 as NA
tiego_raw <- read.csv(tiego_raw_name, na.strings = -999)
# Looks like the CBCL items also have some other missing codes
```

```
tiego_raw |>
    select(starts_with("CBCLProb")) |>
    glimpse()
# Recode 9 to NA as well
tiego_raw <- tiego_raw |>
    mutate(across(starts_with("CBCLProb"),
                  ~ ifelse(.x == 9, NA, .x)))
```

The data contain items from CBCL (Child Behavior Checklist). The items are rated on a 3-point scale (0 = not true, 1 = somewhat or sometimes true, 2 = very true or often true), and should not be treated as continuous. We will focus on the 7-item DSM-oriented Attention Deficit/Hyperactivity Problems scale, with the seven items being Items 4, 8, 10, 41, 78, 93, 104.

```
# Gather the items into a smaller data set
tiego_adhd <- tiego_raw |>
    select(
        adhd1 = CBCLProb4,  # fails to finish
        adhd2 = CBCLProb8,  # concentrate
        adhd3 = CBCLProb10,  # sit still
        adhd4 = CBCLProb41,  # impulsive
        adhd5 = CBCLProb78,  # inattentive
        adhd6 = CBCLProb93,  # talk much
        adhd7 = CBCLProb104  # loud
    ) |>
    # recode to be ordinal
    mutate(
        across(
            everything(),
            ~ ordered(.x, levels = c(0, 1, 2),
                      labels = c("not true", "somewhat or sometimes true",
                                 "very true or often true")),
            .names = "{.col}_c"))
datasummary_skim(tiego_adhd[, 8:14])  # check the recoded items
```

		N	%
adhd1_c	not true	56	41.2
	somewhat or sometimes true	57	41.9
	very true or often true	21	15.4
adhd2_c	not true	74	54.4
	somewhat or sometimes true	39	28.7
	very true or often true	21	15.4
adhd3_c	not true	80	58.8

		N	%
	somewhat or sometimes true	41	30.1
	very true or often true	12	8.8
adhd4_c	not true	80	58.8
	somewhat or sometimes true	38	27.9
	very true or often true	15	11.0
adhd5_c	not true	78	57.4
	somewhat or sometimes true	36	26.5
	very true or often true	20	14.7
adhd6_c	not true	89	65.4
	somewhat or sometimes true	35	25.7
	very true or often true	10	7.4
adhd7_c	not true	103	75.7
	somewhat or sometimes true	24	17.6
	very true or often true	6	4.4

Background on Categorical Data

Many scales in psychology, like the CBCL, use Likert-type response formats (e.g., 0-1-2). Standard CFA using Maximum Likelihood (ML) assumes that the observed variables are:

1. **Continuous**: Can take any value within a range.
2. **Normally Distributed**: Follow a bell curve.

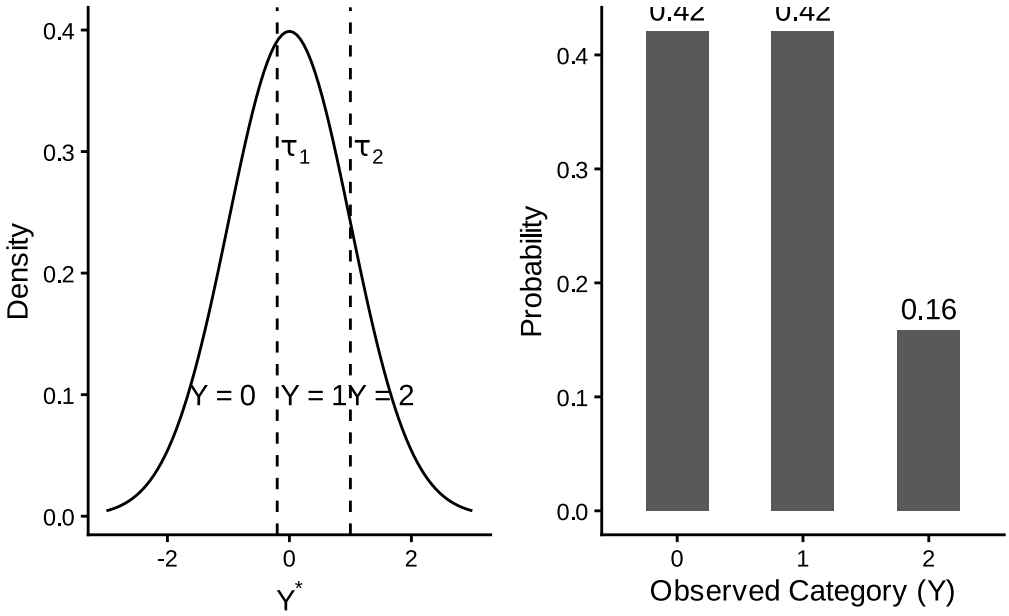
When we have ordinal data with few categories (e.g., fewer than 5), treating them as continuous can lead to:

- Biased parameter estimates (e.g., underestimated factor loadings).
- Incorrect standard errors and χ^2 statistics.
- Spurious multidimensionality (factors defined by item difficulty rather than content).

Underlying Latent Response Formulation

Instead of analyzing the discrete scores directly (0, 1, 2), we assume that for each observed categorical item Y, there is an underlying, unobserved continuous variable Y^* (Y-star). The observed score is determined by where Y^* falls relative to certain **thresholds** (τ):

$$Y = \begin{cases} 0 \text{ if } & Y^* \leq \tau_1 \\ 1 \text{ if } & \tau_1 < Y^* \leq \tau_2 \\ 2 \text{ if } & Y^* > \tau_2 \end{cases}$$



Polychoric Correlations

Because we care about the relationships between the underlying latent responses (Y^*), we estimate the correlations between these Y^* variables rather than the Pearson correlations of the observed integers. These correlations are called **polychoric correlations**. They are typically stronger than the Pearson correlations calculated on coarse categorical data.

WLSMV Estimator

For categorical CFA, we typically use the **WLSMV** (Weighted Least Squares Mean and Variance adjusted) estimator.

- **DWLS**: The core estimation method is Diagonally Weighted Least Squares (DWLS). Unlike Full WLS which requires the inversion of a massive weight matrix (dimensions increase rapidly with more items), DWLS only uses the diagonal elements of the weight matrix. This makes it much more computationally stable and feasible for larger models and smaller sample sizes.
- **MV**: It then applies robust corrections to the standard errors and test statistics (like χ^2) to account for the fact that the full weight matrix was not used. Specifically, it adjusts both the Mean and Variance of the test statistic distribution.

(Not Recommended) Treating Items as Continuous

```
fit_cont <- cfa(
    " ADHD =~ adhd1 + adhd2 + adhd3 + adhd4 + adhd5 + adhd6 + adhd7 ",
    data = tiego_adhd, missing = "fiml", estimator = "MLR"
)
```

```
Warning: lavaan->lav_data_full():
   some cases are empty and will be ignored: 83 132.
```

```
summary(fit_cont, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 32 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        21

                                                   Used       Total
  Number of observations                           134         136
  Number of missing patterns                         4

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                                61.701      50.338
  Degrees of freedom                                14          14
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.226
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                               433.846     336.246
  Degrees of freedom                                21          21
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.290

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.884       0.885
  Tucker-Lewis Index (TLI)                       0.827       0.827

  Robust Comparative Fit Index (CFI)                         0.889
  Robust Tucker-Lewis Index (TLI)                            0.833

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)               -763.251    -763.251
  Scaling correction factor                                  1.143
      for the MLR correction
  Loglikelihood unrestricted model (H1)       -732.400    -732.400
  Scaling correction factor                                  1.176
      for the MLR correction
```

```
  Akaike (AIC)                                1568.501    1568.501
  Bayesian (BIC)                              1629.356    1629.356
  Sample-size adjusted Bayesian (SABIC)       1562.928    1562.928

Root Mean Square Error of Approximation:

  RMSEA                                          0.159       0.139
  90 Percent confidence interval - lower         0.120       0.103
  90 Percent confidence interval - upper         0.201       0.177
  P-value H_0: RMSEA <= 0.050                    0.000       0.000
  P-value H_0: RMSEA >= 0.080                    0.999       0.995

  Robust RMSEA                                               0.156
  90 Percent confidence interval - lower                     0.112
  90 Percent confidence interval - upper                     0.202
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.997

Standardized Root Mean Square Residual:

  SRMR                                           0.074       0.074

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  ADHD =~
    adhd1             1.000                               0.481    0.676
    adhd2             1.258    0.137    9.218    0.000    0.605    0.814
    adhd3             0.794    0.140    5.665    0.000    0.382    0.582
    adhd4             1.115    0.165    6.738    0.000    0.536    0.779
    adhd5             1.354    0.154    8.819    0.000    0.651    0.883
    adhd6             0.640    0.172    3.724    0.000    0.308    0.492
    adhd7             0.522    0.158    3.300    0.001    0.251    0.468

Intercepts:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .adhd1             0.739    0.061   12.018    0.000    0.739    1.038
   .adhd2             0.604    0.064    9.414    0.000    0.604    0.813
   .adhd3             0.486    0.057    8.570    0.000    0.486    0.741
   .adhd4             0.511    0.060    8.581    0.000    0.511    0.742
   .adhd5             0.567    0.064    8.902    0.000    0.567    0.769
   .adhd6             0.410    0.054    7.596    0.000    0.410    0.656
   .adhd7             0.271    0.047    5.831    0.000    0.271    0.506
```

```
Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .adhd1             0.275    0.039    7.002    0.000    0.275    0.543
   .adhd2             0.186    0.034    5.528    0.000    0.186    0.337
   .adhd3             0.284    0.045    6.294    0.000    0.284    0.661
   .adhd4             0.186    0.034    5.440    0.000    0.186    0.393
   .adhd5             0.120    0.029    4.163    0.000    0.120    0.220
   .adhd6             0.296    0.041    7.236    0.000    0.296    0.758
   .adhd7             0.224    0.033    6.729    0.000    0.224    0.781
    ADHD              0.231    0.052    4.451    0.000    1.000    1.000
```

Ordinal CFA

The ordinal CFA approach is based on the polychoric correlations among the items:

```
psych::polychoric(tiego_adhd[, 1:7])
```

```
Call: psych::polychoric(x = tiego_adhd[, 1:7])
Polychoric correlations
      adhd1 adhd2 adhd3 adhd4 adhd5 adhd6 adhd7
adhd1 1.00
adhd2 0.80  1.00
adhd3 0.46  0.61  1.00
adhd4 0.57  0.72  0.57  1.00
adhd5 0.73  0.86  0.61  0.84  1.00
adhd6 0.33  0.37  0.55  0.60  0.55  1.00
adhd7 0.28  0.42  0.55  0.66  0.48  0.65  1.00

 with tau of
          1   2
adhd1 -0.21 1.0
adhd2  0.13 1.0
adhd3  0.26 1.3
adhd4  0.26 1.2
adhd5  0.21 1.0
adhd6  0.42 1.4
adhd7  0.75 1.7
```

```
fit_ord <- cfa(
     " ADHD =~ adhd1_c + adhd2_c + adhd3_c + adhd4_c + adhd5_c + adhd6_c +
adhd7_c ",
    data = tiego_adhd, estimator = "WLSMV", ordered = TRUE
)
summary(fit_ord, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 17 iterations

  Estimator                                       DWLS
  Optimization method                           NLMINB
  Number of model parameters                        21

                                                   Used       Total
  Number of observations                            131         136

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                                34.438      53.338
  Degrees of freedom                                14          14
  P-value (Unknown)                                 NA       0.000
  Scaling correction factor                                  0.687
  Shift parameter                                            3.194
    simple second-order correction

Model Test Baseline Model:

  Test statistic                              1686.338    1026.339
  Degrees of freedom                                21          21
  P-value                                           NA       0.000
  Scaling correction factor                                  1.656

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.988       0.961
  Tucker-Lewis Index (TLI)                       0.982       0.941

  Robust Comparative Fit Index (CFI)                         0.849
  Robust Tucker-Lewis Index (TLI)                            0.774

Root Mean Square Error of Approximation:

  RMSEA                                          0.106       0.147
  90 Percent confidence interval - lower         0.062       0.106
  90 Percent confidence interval - upper         0.151       0.190
  P-value H_0: RMSEA <= 0.050                    0.022       0.000
  P-value H_0: RMSEA >= 0.080                    0.847       0.996

  Robust RMSEA                                               0.247
  90 Percent confidence interval - lower                     0.164
  90 Percent confidence interval - upper                     0.333
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.999

Standardized Root Mean Square Residual:
```

```
  SRMR                                           0.101       0.101

Parameter Estimates:

  Parameterization                               Delta
  Standard errors                           Robust.sem
  Information                                 Expected
  Information saturated (h1) model        Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  ADHD =~
    adhd1_c           1.000                               0.772    0.772
    adhd2_c           1.178    0.101   11.679    0.000    0.910    0.910
    adhd3_c           0.889    0.096    9.300    0.000    0.686    0.686
    adhd4_c           1.125    0.089   12.580    0.000    0.868    0.868
    adhd5_c           1.213    0.089   13.674    0.000    0.937    0.937
    adhd6_c           0.832    0.115    7.256    0.000    0.642    0.642
    adhd7_c           0.826    0.116    7.119    0.000    0.638    0.638

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    adhd1_c|t1       -0.202    0.111   -1.827    0.068   -0.202   -0.202
    adhd1_c|t2        0.993    0.132    7.519    0.000    0.993    0.993
    adhd2_c|t1        0.144    0.110    1.305    0.192    0.144    0.144
    adhd2_c|t2        0.993    0.132    7.519    0.000    0.993    0.993
    adhd3_c|t1        0.281    0.112    2.521    0.012    0.281    0.281
    adhd3_c|t2        1.331    0.154    8.655    0.000    1.331    1.331
    adhd4_c|t1        0.242    0.111    2.174    0.030    0.242    0.242
    adhd4_c|t2        1.203    0.144    8.335    0.000    1.203    1.203
    adhd5_c|t1        0.202    0.111    1.827    0.068    0.202    0.202
    adhd5_c|t2        1.025    0.134    7.666    0.000    1.025    1.025
    adhd6_c|t1        0.403    0.113    3.558    0.000    0.403    0.403
    adhd6_c|t2        1.430    0.162    8.811    0.000    1.430    1.430
    adhd7_c|t1        0.742    0.122    6.100    0.000    0.742    0.742
    adhd7_c|t2        1.687    0.191    8.846    0.000    1.687    1.687

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .adhd1_c           0.404                               0.404    0.404
   .adhd2_c           0.173                               0.173    0.173
   .adhd3_c           0.529                               0.529    0.529
   .adhd4_c           0.246                               0.246    0.246
   .adhd5_c           0.123                               0.123    0.123
   .adhd6_c           0.587                               0.587    0.587
   .adhd7_c           0.593                               0.593    0.593
    ADHD              0.596    0.081    7.396    0.000    1.000    1.000
```

The fit indices are different from those in the continuous CFA, and does not appear adequate. We can check the residuals and modification indices to see where the model misfits.

```
resid(fit_ord, type = "cor.bentler")$cov
```

```
        adhd1_ adhd2_ adhd3_ adhd4_ adhd5_ adhd6_ adhd7_
adhd1_c  0.000
adhd2_c  0.097  0.000
adhd3_c -0.071 -0.018  0.000
adhd4_c -0.100 -0.069 -0.019  0.000
adhd5_c  0.003  0.008 -0.024  0.029  0.000
adhd6_c -0.162 -0.209  0.119  0.034 -0.051  0.000
adhd7_c -0.210 -0.156  0.118  0.105 -0.119  0.240  0.000
```

```
modindices(fit_ord, sort. = TRUE, minimum.value = 10)
```

```
       lhs op     rhs     mi  epc sepc.lv sepc.all sepc.nox
45 adhd1_c ~~ adhd2_c 11.205 0.26    0.26    0.984    0.984
```

Updated Models

1. Add unique covariance between `adhd1_c` and `adhd2_c` and between adhd6_c and adhd7_c based on modification indices.

> **Reminder**
>
> When adding unique covariances, make sure they make theoretical sense. Also, avoid adding too many unique covariances post hoc.

```
fit_ord2 <- cfa(
     " ADHD =~ adhd1_c + adhd2_c + adhd3_c + adhd4_c + adhd5_c + adhd6_c
+ adhd7_c
      adhd1_c ~~ adhd2_c
      adhd6_c ~~ adhd7_c ",
    data = tiego_adhd, estimator = "WLSMV", ordered = TRUE
)
summary(fit_ord2, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 21 iterations

  Estimator                                       DWLS
  Optimization method                           NLMINB
  Number of model parameters                        23
```

```
                                                   Used       Total
  Number of observations                            131         136

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                                14.788      27.170
  Degrees of freedom                                12          12
  P-value (Unknown)                                 NA       0.007
  Scaling correction factor                                  0.604
  Shift parameter                                            2.673
    simple second-order correction

Model Test Baseline Model:

  Test statistic                              1686.338    1026.339
  Degrees of freedom                                21          21
  P-value                                           NA       0.000
  Scaling correction factor                                  1.656

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.998       0.985
  Tucker-Lewis Index (TLI)                       0.997       0.974

  Robust Comparative Fit Index (CFI)                         0.938
  Robust Tucker-Lewis Index (TLI)                            0.891

Root Mean Square Error of Approximation:

  RMSEA                                          0.042       0.099
  90 Percent confidence interval - lower         0.000       0.049
  90 Percent confidence interval - upper         0.104       0.148
  P-value H_0: RMSEA <= 0.050                    0.521       0.053
  P-value H_0: RMSEA >= 0.080                    0.186       0.762

  Robust RMSEA                                               0.171
  90 Percent confidence interval - lower                     0.055
  90 Percent confidence interval - upper                     0.276
  P-value H_0: Robust RMSEA <= 0.050                         0.045
  P-value H_0: Robust RMSEA >= 0.080                         0.918

Standardized Root Mean Square Residual:

  SRMR                                           0.067       0.067

Parameter Estimates:

  Parameterization                               Delta
```

```
  Standard errors                            Robust.sem
  Information                                  Expected
  Information saturated (h1) model          Unstructured

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  ADHD =~
    adhd1_c           1.000                               0.670    0.670
    adhd2_c           1.260    0.126    9.991    0.000    0.844    0.844
    adhd3_c           1.053    0.125    8.394    0.000    0.706    0.706
    adhd4_c           1.316    0.134    9.848    0.000    0.882    0.882
    adhd5_c           1.451    0.155    9.375    0.000    0.973    0.973
    adhd6_c           0.896    0.146    6.142    0.000    0.601    0.601
    adhd7_c           0.897    0.156    5.766    0.000    0.601    0.601

Covariances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
 .adhd1_c ~~
   .adhd2_c           0.234    0.066    3.551    0.000    0.234    0.588
 .adhd6_c ~~
   .adhd7_c           0.289    0.097    2.975    0.003    0.289    0.452

Thresholds:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
    adhd1_c|t1       -0.202    0.111   -1.827    0.068   -0.202   -0.202
    adhd1_c|t2        0.993    0.132    7.519    0.000    0.993    0.993
    adhd2_c|t1        0.144    0.110    1.305    0.192    0.144    0.144
    adhd2_c|t2        0.993    0.132    7.519    0.000    0.993    0.993
    adhd3_c|t1        0.281    0.112    2.521    0.012    0.281    0.281
    adhd3_c|t2        1.331    0.154    8.655    0.000    1.331    1.331
    adhd4_c|t1        0.242    0.111    2.174    0.030    0.242    0.242
    adhd4_c|t2        1.203    0.144    8.335    0.000    1.203    1.203
    adhd5_c|t1        0.202    0.111    1.827    0.068    0.202    0.202
    adhd5_c|t2        1.025    0.134    7.666    0.000    1.025    1.025
    adhd6_c|t1        0.403    0.113    3.558    0.000    0.403    0.403
    adhd6_c|t2        1.430    0.162    8.811    0.000    1.430    1.430
    adhd7_c|t1        0.742    0.122    6.100    0.000    0.742    0.742
    adhd7_c|t2        1.687    0.191    8.846    0.000    1.687    1.687

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .adhd1_c           0.551                               0.551    0.551
   .adhd2_c           0.287                               0.287    0.287
   .adhd3_c           0.502                               0.502    0.502
   .adhd4_c           0.222                               0.222    0.222
   .adhd5_c           0.054                               0.054    0.054
   .adhd6_c           0.639                               0.639    0.639
```

```
   .adhd7_c          0.639                          0.639    0.639
    ADHD             0.449    0.089    5.037    0.000    1.000    1.000
```